

Banking Customer Churn Analytics Project Report

Executive Summary

This project analyzes a banking customer churn dataset to identify customer characteristics, behavioural patterns, financial factors, product usage, geographic trends, and customer segments associated with customer churn.

The analysis was performed using **Python for data cleaning and exploratory data analysis (EDA)**, **SQL Server for business analysis through 67 analytical queries**, and **Power BI to develop seven interactive dashboards** that enable banking stakeholders to monitor customer churn, identify high-risk customer segments, and support data-driven retention strategies.

The analysis focuses on understanding why customers leave the bank, which customer groups experience higher churn, how customer engagement and banking-product usage relate to retention, and which areas require targeted intervention.

Key Findings

1. Executive Churn Overview

Findings

- The dataset contains **10,000 customers**.
- **2,037 customers have churned**, representing an overall churn rate of **20.37%**.
- The majority of customers have been retained, but approximately one in five customers has exited.
- Customer activity is strongly associated with retention behaviour.
- Customer churn varies considerably across geographic markets.
- Customer age, product usage, and financial characteristics show different levels of association with churn.

Business Recommendations

- Establish customer churn as a key performance indicator for ongoing monitoring.
- Develop targeted retention strategies for high-churn customer segments.
- Monitor customer activity levels as an early indicator of potential churn.
- Investigate geographic markets with significantly higher churn rates.
- Use Power BI dashboards to continuously monitor customer retention performance.

2. Customer Demographics & Churn

Findings

- Customer churn varies significantly across geographic markets.
- **Germany records the highest churn rate among the three geographies**, making it an important market for further investigation.
- Churn also differs between male and female customers.
- Customer age is an important characteristic when analysing churn behaviour.
- Older customer groups generally demonstrate higher churn levels than younger customer groups.
- Customer tenure provides additional insight into differences in retention behaviour.

Business Recommendations

- Investigate the underlying causes of higher churn in Germany.

- Develop market-specific retention strategies rather than applying a single strategy across all geographies.
- Analyse customer needs across different age groups.
- Develop targeted engagement campaigns for age groups with elevated churn.
- Monitor churn behaviour across customer tenure groups to identify vulnerable stages in the customer relationship.

3. Banking Product & Customer Engagement Analysis

Findings

- Customer churn varies considerably according to the number of banking products held.
- Customers with certain product counts demonstrate substantially higher churn rates than other product groups.
- Customer activity is strongly associated with retention.
- **Inactive customers demonstrate considerably higher churn than active customers.**
- Credit-card ownership provides an additional dimension for understanding customer behaviour.
- Product adoption and customer engagement therefore represent important areas for retention analysis.

Business Recommendations

- Prioritize inactive customers for customer-engagement and reactivation campaigns.
- Investigate why customers with particular product combinations experience elevated churn.
- Encourage appropriate cross-selling and product adoption where it can strengthen customer relationships.
- Use customer activity as an early-warning indicator for potential churn.
- Develop personalised product recommendations based on customer needs and behaviour.
- Monitor product engagement regularly to identify declining customer relationships.

4. Financial & Credit Profile Analysis

Findings

- Customer churn varies across different credit-score bands.
- Account balances also differ between churned and retained customers.
- Customers with different balance levels demonstrate different churn patterns.
- Estimated salary provides another dimension for analysing customer churn.
- Financial characteristics can help identify customer segments that require closer monitoring.
- Credit score, account balance, and estimated salary should be considered together rather than independently when evaluating customer behaviour.

Business Recommendations

- Develop targeted retention strategies for financially significant customer segments.
- Investigate customers whose financial profiles are associated with elevated churn.
- Use customer financial characteristics to support personalised engagement.
- Avoid relying on a single financial indicator when assessing churn risk.
- Combine financial information with customer activity and product usage to improve customer segmentation.

5. Geography & Customer Behaviour Analysis

Findings

- Customer churn differs substantially across geographic markets.
- Germany represents the highest-churn geography in the dataset.
- Churn patterns also vary when geography is combined with gender.
- Customer activity levels differ in their relationship with churn across geographic markets.
- The combination of geography and number of products reveals additional high-churn customer segments.
- Geographic analysis demonstrates that customer retention challenges are not evenly distributed across markets.

Business Recommendations

- Conduct a deeper investigation into customer experience in high-churn geographic markets.
- Develop market-specific customer retention strategies.
- Compare customer engagement levels across geographic markets.
- Investigate high-churn geography/product combinations.
- Allocate retention resources according to customer risk and market-specific churn patterns.
- Monitor geographic churn trends regularly through Power BI reporting.

6. Customer Risk Analysis

Findings

- Inactive customers represent an important customer group for churn prevention.
- Customers who are both inactive and churned represent a significant opportunity for understanding customer disengagement.
- Churn varies across combinations of age group, activity status, and number of products.
- Certain combinations of customer characteristics demonstrate substantially higher churn rates than the overall benchmark.
- A simple customer-risk segmentation was developed to help distinguish customers according to their churn and engagement status.

Business Recommendations

- Prioritize inactive customers for proactive engagement before they become churned customers.
- Develop early-warning monitoring based on declining customer activity.
- Investigate high-churn combinations of age, activity, geography, and product usage.
- Create targeted retention campaigns for elevated-risk customer segments.
- Regularly update customer segmentation as customer behaviour changes.
- Use predictive analytics in future iterations to develop more sophisticated churn-risk models.

7. Retention Strategy & Recommendations

Findings

- Customer churn represents a significant retention challenge, with an overall churn rate of **20.37%**.
- Customer activity is one of the clearest behavioural dimensions associated with churn.
- Geographic markets show different levels of customer retention.
- Product usage is associated with different churn patterns.

- Customer demographics and financial characteristics provide additional information for identifying customer segments that require attention.
- A combination of customer characteristics provides a more useful view of churn than analysing individual variables in isolation.

Business Recommendations

- Develop proactive retention programmes for inactive customers.
- Investigate the causes of elevated churn in Germany.
- Develop targeted retention campaigns for high-churn demographic and product segments.
- Strengthen customer engagement through personalised communication and relevant banking-product offerings.
- Monitor high-risk customer segments continuously.
- Use predictive analytics to identify customers who are likely to churn before they leave.

Overall Business Insights

The analysis demonstrates that customer churn is influenced by a combination of **customer demographics, geographic location, customer activity, banking-product usage, and financial characteristics**.

Customer activity is particularly important because inactive customers demonstrate substantially higher churn behaviour than active customers.

Geographic differences also represent an important area of concern, with Germany recording the highest churn rate among the three markets analysed.

Product usage provides another important dimension. Customers with different numbers of banking products demonstrate different churn patterns, suggesting that product engagement and the depth of the customer relationship may influence retention.

Financial characteristics such as credit score, account balance, and estimated salary provide additional information that can be used to understand customer segments and support more targeted retention strategies.

Overall, the analysis demonstrates the importance of moving beyond simply measuring the number of customers who churn. A bank can gain greater value by identifying **who is churning, where they are located, how engaged they are, what products they use, and what characteristics are associated with elevated churn**.

Overall Recommendations

- **Prioritize inactive customers** through proactive engagement and reactivation campaigns.
- **Investigate Germany's elevated churn rate** and identify potential customer-experience or market-specific causes.
- **Strengthen customer engagement** through personalised communication and relevant banking-product offerings.
- **Monitor product usage** and identify customer segments with unusually high churn.
- **Develop targeted retention strategies** based on geography, age, activity, product usage, and financial characteristics.
- **Establish customer churn as a continuous KPI** for management monitoring.
- **Use Power BI dashboards** to provide management with continuous visibility into customer retention patterns.
- **Introduce predictive churn modelling** in future phases to identify customers who are likely to leave before they actually churn.
- **Combine behavioural, demographic, product, and financial data** when assessing customer retention opportunities.

- **Continuously evaluate retention campaigns** to determine which strategies produce the strongest improvement in customer retention.

Conclusion

The Banking Customer Churn Analytics project demonstrates how an end-to-end data analytics workflow can transform raw customer data into meaningful banking insights.

By integrating **Python for exploratory data analysis, SQL Server for structured business analysis, and Power BI for interactive reporting**, the project provides a comprehensive view of customer churn and the factors associated with customer retention.

The analysis provides banking stakeholders with a framework for identifying high-churn customer segments, understanding customer behaviour, monitoring retention performance, and developing targeted strategies to reduce customer attrition.

Ultimately, the project demonstrates how **data-driven customer retention analytics can support better decision-making, improve customer engagement, and help financial institutions protect long-term customer relationships.**